



Svala Sverrisdóttir
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EDUCATION

University of California at Berkeley

2022-2026

Doctorate of Philosophy in Mathematics, United States of America

Advised by Bernd Sturmfels, with a focus on computational algebraic geometry, combinatorics, and commutative algebra with applications in physics and quantum chemistry. Graduate spring 2026 with a thesis titled *Non linear algebra in quantum chemistry*.

University of Iceland

2019 - 2022

Bachelor of Science in Mathematics, Iceland

Focus on applied mathematics.

GPA is **9.95/10**, first-class grade with distinction.

EXPERIENCE

Researcher at the Max Planck Institute for Mathematics in the Sciences, Leipzig Germany

2022 - present

Full-time researcher for the 2025-2026 academic year and during summers, in the non-linear algebra group led by MPI director Bernd Sturmfels.

Graduate Student Instructor at UC Berkeley

2023 - present

Taught three courses: Linear Algebra and Differential Equations, Calculus, and Introduction to Abstract Algebra.

Software Developer at PayAnalytics

2020 - 2022

Full-time during summers, part-time along studies, at a Reykjavík based startup, which was recently acquired by Beqom. Main responsibilities included programming for the frontend in JavaScript (Vue.js), backend in Python (Django) and statistical reporting for clients.

Undergraduate Student Instructor at the University of Iceland

2020-2021

Taught three courses: Probability and Statistics, Honors Linear Algebra, and Mathematical Analysis I.

RESEARCH

Algebraic Varieties in Quantum Chemistry

Cited by 12

Fabian Faulstich, Bernd Sturmfels and Svala Sverrisdottir,
Foundations of Computational Mathematics (2024) 1-32. (arXiv)

Coupled Cluster Degree of the Grassmannian

Cited by 7

Viktoriia Borovik, Bernd Sturmfels and Svala Sverrisdottir,
Journal of Symbolic Computation 128 (2025) 102396. (arXiv)

Exploring Ground and Excited States via Single Reference CC Theory and Algebraic Geom.

Cited by 4

Fabian Faulstich and Svala Sverrisdottir,
Journal of Chemical Theory and Computation 20 (2024) 8517-8528. (arXiv)

Kinematic Varieties for Massless Particles

Cited by 2

Smita Rajan, Bernd Sturmfels and Svala Sverrisdottir,
Le Matematiche 80 (2025) 453-471. (arXiv)

Gram Matrices for Isotropic Vectors

Cited by 1

Yassine El Maazouz, Bernd Sturmfels and Svala Sverrisdottir,
submitted to *Communications in Mathematical Physics*. (arXiv)

Algebraic Varieties in Second Quantization

Svala Sverrisdottir,
to appear in *SIAM Journal on Applied Algebra and Geometry*. (arXiv)

Four-Dimensional Lie Algebras Revisited

Laurent Manivel, Bernd Sturmfels and Svala Sverrisdottir,
Journal of Lie Theory 33 (2023) 703-712. (arXiv)

TECHNICAL SKILLS

Languages: English (fluent), Icelandic (fluent), Danish (conversational).

Programming: Julia, Python, Java, Javascript, Macaulay2.

ACHIEVEMENTS AND AWARDS

The Leifur Eiriksson Foundation Fellowship Funds scholars who are Icelandic citizens or permanent residents from universities in Iceland to conduct graduate research or study at universities in the United States.	<i>2025- 2026</i>
The Soroptimist International of Europe Scholarship A scholarship that helps to educate, empower and enable women and girls.	<i>Fall 2025</i>
Icelandic Chamber of Commerce Educational Grant To support Icelandic students studying abroad and enhance the competitiveness of the Icelandic economy in the global market.	<i>Spring 2025</i>
Simons Fellowship in Math One semester fellowship from the University of California, Berkeley.	<i>Spring 2025</i>
Memorial fund of Helga Jónsdóttir and Sigurliði Kristjánsson To support students in engineering and science studies.	<i>Fall 2024</i>
Chancellors Fellowship One semester fellowship from the University of California, Berkeley.	<i>Fall 2022</i>
Sigurður Helgason Award Fund Awarded annually to the top student(s) in the mathematics department at the University of Iceland.	<i>Fall 2021</i>

OUTREACH AND ORGANIZATION

Co-organizer for the IPAM workshop Algebraic varieties and electronic structure A workshop at the Institute for Pure and Applied Mathematics (IPAM) long-term program for Spring 2027, titled Numerical Algebraic Geometry and Correlated Electrons: Generalized Grassmannians Response Functions and Excited States.	<i>Spring 2027</i>
Stelpur diffra! (Girls differentiate!) - Math camp for teen girls Hosted an abstract algebra workshop introducing high school girls and teachers to college-level algebra. We study, for example, the dihedral groups, symmetric groups, and Burnside's lemma.	<i>August 2024</i>
Website development - stigull.is Developed the backend and the frontend for the math student organization website at the University of Iceland. Included code for creating an account, logging in/out and signing up for events.	<i>2021</i>

RESEARCH TALKS

Gram Matrices for Isotropic Vectors Cosmology Meets Non-Linear Algebra Workshop at the University of Amsterdam	<i>Nov. 3rd 2025</i>
Algebraic Varieties in Second Quantization Applied CATS seminar at KTH in Stockholm	<i>Oct. 14th 2025</i>
Algebraic Varieties in Second Quantization TENORS Workshop I at the University of Konstanz	<i>Sept. 25th 2025</i>
Ground and Excited States in Coupled Cluster Theory Algebra and Geometry of the Quantum Many-body Problem Minisymposium at SIAM AG25	<i>July 7th 2025</i>
Algebraic Varieties in Second Quantization Information, Geometry and Physics seminar in Cal-Tech.	<i>Apr. 30th 2025</i>
Algebraic Varieties in Second Quantization Math Frontier Seminar at RPI in Troy.	<i>Mar. 31st 2025</i>
Algebraic Varieties in Second Quantization Algebra and Discrete Math Seminar and UC Davis.	<i>Feb. 25th 2025</i>
Algebraic Varieties in Second Quantization Quantum Lunch at the Centre for the Mathematics of Quantum Theory, University of Copenhagen.	<i>Feb. 12th 2025</i>
Gram matrices for Isotropic Vectors AMS Special Session on Applications of Algebraic Geometry at the JMM 2025.	<i>Jan. 11th 2025</i>
Gram matrices for Isotropic Vectors Nonlinear Algebra Seminar at UC Berkeley.	<i>Dec. 12th 2024</i>
Algebraic Varieties Arising from Quantum Chemistry Optimization Techniques in Quantum Chemistry Workshop at RWTH Aachen University.	<i>June 6th 2024</i>
Toric Degenerations Arising from Quantum Chemistry Seminar on Nonlinear Algebra at MPI for Mathematics in the Sciences, Leipzig.	<i>May 28th 2024</i>
CC Degree of the Grassmannian Special Session on Research in Combinatorics by Early Career Mathematicians at the AMS Sectional Meeting at SFSU, San Francisco.	<i>May 4th 2024</i>
Four-Dimensional Lie Algebras Revisited Oberseminar Algebra and Discrete Math at University of Osnabrück.	<i>July 4th 2023</i>